

HSV Suppression in People Living with HIV

For **valaciclovir**, multiple studies have evaluated its role in suppressing recurrent genital herpes. High-dose regimens (2 g four times daily) have been investigated in immunocompromised populations, including those with HIV and individuals recovering from bone marrow transplants. More recently, numerous trials have assessed the efficacy of **aciclovir** and **valaciclovir** for HSV suppression and their impact on HIV transmission in co-infected individuals. These studies show that oral **aciclovir** at standard doses, as well as **valaciclovir** at 1 g once daily or 500 mg twice daily, are associated with minimal to no adverse effects or toxicity compared to HIV-negative individuals. However, **high-dose valaciclovir (8 g daily)** has been linked to **microangiopathic haemolytic uraemic syndrome**.

Recommended Regimens for Daily HSV Suppressive Therapy in HIV-Positive Individuals:

- **Aciclovir 400 mg** orally twice to three times a day
- **Valaciclovir 500 mg** orally twice a day

If these regimens fail to control outbreaks, the first step is to **double the dose**. If suppression remains inadequate, **famciclovir 500 mg** orally twice a day may be considered (evidence level: 1, B).

HSV Suppression and HIV Progression

Suppressive antiviral therapy with **aciclovir** or **valaciclovir** has been shown to reduce HIV viraemia in patients with detectable viral loads, although the exact mechanism remains unclear. This effect can influence HIV disease progression, particularly in individuals **not on antiretroviral therapy (ART)**.

A large randomized controlled trial (RCT) in patients with early HIV (not on ART, CD4 counts >250 cells/mm³) demonstrated that **aciclovir 400 mg**

twice daily helped maintain CD4 counts above 250 cells/mm³, reducing the need for initiating ART by **16%** at 2 years in the treatment group.

However, **no significant benefit** of HSV suppressive therapy has been observed in individuals already receiving effective ART. Since effective ART remains the cornerstone of HIV management—both for individual health and prevention of transmission—**HSV suppression for the purpose of controlling HIV progression is not considered clinically relevant** in the modern treatment era.

Management of Recalcitrant Herpes in Immunocompromised Individuals

Clinically refractory genital herpes, while rare in immunocompetent individuals, is a significant issue in those with severe immunodeficiency, such as in late-stage HIV.

Treatment Approach:

- **Systemic therapy with foscarnet or cidofovir** is preferred for drug-resistant HSV in HIV-positive patients.
- Evidence supports **alternating aciclovir and cidofovir** for recurrent episodes, which may help reduce the emergence of cidofovir-resistant strains.
- However, the **efficacy, safety, and durability** of these treatments require further evaluation in prospective, controlled trials.

Aciclovir Resistance:

- Aciclovir-resistant HSV strains are found in approximately **5–7%** of genital herpes isolates from people with HIV.
- Resistance is confirmed when isolates require **aciclovir concentrations >1–3 mg/L** for inhibition.
- Most commonly, resistance arises from mutations in the **HSV thymidine kinase (TK) gene**, which impairs the initial phosphorylation of aciclovir.

- Mutations may lead to TK with reduced affinity for aciclovir or complete absence of TK production.
- **TK-deficient strains** are less pathogenic in immunocompetent hosts but can cause severe local or disseminated disease in immunocompromised individuals.
- These strains appear less capable of establishing latency; thus, subsequent recurrences are often due to **aciclovir-sensitive isolates**.

Treatment Implications:

- **Partially resistant strains** may respond to high-dose intravenous aciclovir or other nucleoside analogues.
- **Fully resistant strains** are cross-resistant to **valaciclovir** and **ganciclovir**, and most are also resistant to **famciclovir**.